

Madhan Raj C V

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EDUCATION

SRM Institute of Science and Technology

Vadapalani, Chennai

Bachelor of Technology in Computer Science and Business Systems

2021 – 2025

– CGPA: 8.3/10.0

Velammal Vidhyashram

Surapet (CBSE)

12th Grade – 78%

2020

Velammal Vidhyashram

Surapet (CBSE)

10th Grade – 72%

2018

EXPERIENCE

Program Tester (Gen AI Intern)

Chennai, TN

Evolve Robot Lab

- Developed and fine-tuned a Generative AI model for high-accuracy text summarization tasks.
- Implemented and evaluated machine learning classification models using Python and Scikit-learn.
- Gained hands-on experience in full ML lifecycle including data preprocessing, training, and evaluation.

Support Intern

Chennai, TN

CPCL (Chennai Petroleum Corporation Limited)

- Maintained critical IT infrastructure and monitored refinery systems to ensure system uptime.
- Resolved hardware and network issues and documented system workflows with technical team.

PROJECTS

Microservices-Based E-Commerce System | Java, Spring Boot, Spring Cloud, Netflix Eureka, MySQL

- Architected a scalable E-commerce application featuring decoupled microservices for Product Catalog and Order Management.
- Integrated Spring Cloud and Netflix Eureka for dynamic service registration, discovery, and inter-service communication.
- Developed secure RESTful APIs for inventory tracking, using Java Stream API for highly efficient catalog filtering.
- Integrated Spring Data JPA for persistence with a MySQL database, processing complex nested JSON payloads for orders.
- Automated API contract documentation using Swagger UI, enabling validation of checkout endpoints and edge cases.

Employee Management System with Spring JPA Data | Java, Spring Boot, Spring Data JPA, MySQL

- Developed a backend data infrastructure using Spring Boot, managing complex database tables ('emp_details', 'Addr', 'Parking', 'Department').
- Implemented diverse relational mappings using Hibernate, comprising **One-to-One** ('AddrDetails'), **One-to-Many** with Eager fetching ('List{Parking}'), and **Many-to-One** relationships.
- Designed optimized primary key generation routines using custom database '**SequenceGenerator**' models for thread-safe entity allocation.
- Configured flexible query capabilities within 'JpaRepository', establishing Spring Data Derived Query Methods alongside custom **Native SQL** and **JPQL** query expressions using parameterized binding ('@Param').
- Utilized Spring's ApplicationContext container to initialize application runtimes and handle transactional enterprise operations smoothly.

TECHNICAL SKILLS

Languages: Java (Stream API, Collections, Multithreading), Python, SQL (MySQL), JavaScript, HTML5, CSS3

Frameworks: Spring, Spring Boot (Web, JPA, JDBC, Cloud, Netflix Eureka)

Tools: Swagger UI, Postman, Spring Tool Suite, IntelliJ IDEA, VS Code, MySQL Workbench

Software Development: Microservices Architecture, RESTful API Design, MVC Architecture, Database Normalization